

ANJANAA S

AI/ML Engineer | AI Researcher

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PROFESSIONAL SUMMARY

AI/ML Engineer and Researcher with expertise in Applied AI, Machine Learning, Generative AI, and Predictive Modeling. Experienced in developing intelligent systems, real-time AI pipelines, and evaluation frameworks across research and industry internships, with publications in leading international conferences.

EDUCATION

B.Tech in Computer Science and Medical Engineering **2022 – 2026**

Sri Ramachandra Institute of Higher Education and Research (SRIHER) | **CGPA: 9.5/10**

Higher Secondary Certificate (Bio-Math) – 90% **2021 – 2022**

Swamy's Matric Higher Secondary School, Tamil Nadu

Secondary School Leaving Certificate – 95% **2019 – 2020**

Swamy's Matric Higher Secondary School, Tamil Nadu

EXPERIENCE

Research Intern (AI/ML) | DRDO – Defence Research & Development Organisation **06/2026 – Present**

- Developed a benchmark for evaluating temporal reasoning, causal consistency, and structural invariance in text-to-video generation.
- Built an LLM-driven CER pipeline for automated preference pair generation and reasoning-based QA validation.
- Developed a DPO-based preference optimization framework to improve negation understanding and semantic alignment in text-to-video generation.

Software Developer Intern | ISRO Health Centre, URSC Bangalore **05/2026 – 06/2026**

- Developed a multi-protocol Laboratory Information System (LIS) integration framework for clinical analyzers.
- Implemented TCP/IP, ASTM, and RS232 communication for secure, real-time clinical data exchange.
- Automated analyzer connectivity and laboratory workflows through barcode-enabled sample processing.

AI/ML Engineer | ISRO Bangalore **02/2026 – 04/2026**

- Developed a hybrid Battery Digital Twin for real-time SOC, SOH, and RUL estimation of spacecraft Li-ion batteries.
- Built Flask-SocketIO telemetry pipelines, RS232 integration, and real-time monitoring dashboards.
- Implemented EKF, Arrhenius-Woehler, and LSTM/GRU models for battery degradation prediction and health analytics.

Research Intern | IIITDM Kancheepuram **08/2024 – 11/2024**

- Developed and validated Spatially Offset High-Resolution Microendoscopy (SO-HRME) systems.
- Conducted optical imaging experiments and fiber-bundle contrast degradation analysis.
- Contributed to publications presented at EMBC 2025 (Denmark) and IEEE Photonics IPC 2025 (Singapore).

Research Intern (ML) | SRIHER **Feb 2024 – Apr 2024**

- Built ML-based multiclass classification models for hearing disability detection using PTA audiometry data.
- Worked on real-time audiometry datasets in collaboration with a clinical hearing center.

Summer Intern – Web Development | SRIHER **May 2023 – Jul 2023**

- Developed SRET login and feedback portal using MySQL, PHP, and XAMPP with frontend and backend integration.

RESEARCH PUBLICATIONS & PROJECTS

AI-Powered Emergency Decision Support System for Golden Hour | ICCCN 2026

- Hybrid multimodal AI framework using ViTs, XGBoost, GNNs, and LLMs for trauma severity prediction and intelligent hospital routing.

Multimodal AI for Early Sepsis Prediction | HACK4HEALTH 2026

- Hybrid LSTM, XGBoost, LLM, and Agentic AI framework with SHAP/LIME explainability for early sepsis prediction and clinical decision support.

High-Resolution Microendoscopy (SO-HRME) & Imaging Optimization | EMBC 2025, IEEE Photonics IPC 2025

- Optimized SO-HRME imaging systems by quantifying fiber-bundle bending effects for improved subsurface tissue imaging and early cancer detection.

Reinforcement Learning Framework for Personalized Chemotherapy Decision Support | ICCIS 2025

- RL-based ensemble framework using genomic and clinical data for personalized chemotherapy planning. Best Paper Award – ICCIS 2025.

HyREAL: Hybrid Reinforcement Learning for Survival Prediction | CIMA 2025

- Hybrid RL + ensemble framework achieving 94% accuracy with SHAP-based explainability.

GAN-Based Transfer Learning for Brain Tissue Classification in Progressive MS | SIPCOV 2025

- GAN-augmented VGG16 model achieving 93% accuracy for EM-based brain tissue classification. Best Paper Award – SIPCOV 2025.

SliceNet: Explainable CT-Based Bladder Cancer Stage Classification | AITA 2025

- Lightweight CNN with patch-based Grad-CAM visualization for explainable bladder cancer stage classification.

ACHIEVEMENTS

- 1st Prize – HACK4HEALTH, National Level Hackathon, Anna University
- 1st Prize – AI Tool Buildathon 2026, SRIHER (AI Innovation in Medical Engineering)
- Best Paper Award – ICCIS 2025 (RL Chemotherapy Framework) & SIPCOV 2025 (GAN-based MS Classification)
- Poster Presentation – EMBC 2025, Denmark | Paper Presentation – IEEE Photonics IPC 2025, Singapore
- 2nd Prize – Research Day 2025, SRIHER | NCC Naval Wing Cadet, B & C Certificates
- University Athlete – 100m & 200m Sprint

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, HTML/CSS, JavaScript
- **AI & Machine Learning:** Machine Learning, Deep Learning, Large Language Models (LLMs), Generative AI, Computer Vision, Multimodal AI, Reinforcement Learning, Explainable AI
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Flask, Flask-SocketIO, Pandas, NumPy
- **Models & Algorithms:** Transformers, CNNs, LSTM, GRU, XGBoost, GANs, Random Forest, Extended Kalman Filter (EKF)
- **Systems & Technologies:** Digital Twins, TCP/IP, ASTM, RS232, Socket Programming
- **Visualization & Engineering Tools:** Power BI, Tableau, MATLAB, COMSOL

CERTIFICATIONS & TRAINING

- Python Fundamentals & Foundation of AI – Great Learning (Mar 2023)
- Machine Learning Terminologies & Process – AWS (Feb 2024)
- Intro to Machine Learning: Art of Possible – AWS (Feb 2024)
- Business Intelligence Using Power BI – Skill Nation (Jul 2024)
- Vaccine Technology & Cancer Biology & Technology – FCC, IIT Hyderabad (Jul–Oct 2024)